

Density clustering (DBSCAN / HDBSCAN): tried, not worth pursuing for cell segmentation

We detected the lipid droplets in each image and grouped them into “cells” with DBSCAN and HDBSCAN. We tested a positive image (many fat cells) and a near-empty image (almost none), and swept DBSCAN’s radius ϵ from 10 to 30 px.

- **DBSCAN breaks cells apart or merges them.** With a small radius ($\epsilon = 12$), single cells split into pieces and 65% of droplets are thrown away (188 clusters on the positive image). With a larger radius, touching cells merge into big blobs. No setting does both jobs, and the best setting changes from image to image.
- **HDBSCAN merges neighbouring cells** (55 clusters on the positive image). On the near-empty image, it turned background noise into 2 huge false “clusters”.
- **Root cause:** inside a cell, droplets are about as close together as they are across the edge between two touching cells. The cell edge is a change in brightness, not a gap between droplets, so droplet positions alone can’t separate touching cells. That’s the same problem the current IMES tool has.
- **Verdict:** dropped as a segmentation method. It’s kept only as a simple baseline, for measuring cluster area, and possibly as a source of automatic prompts. We moved on to learned models (Cellpose-SAM, cpdino, micro-SAM), which use the cell edges.

